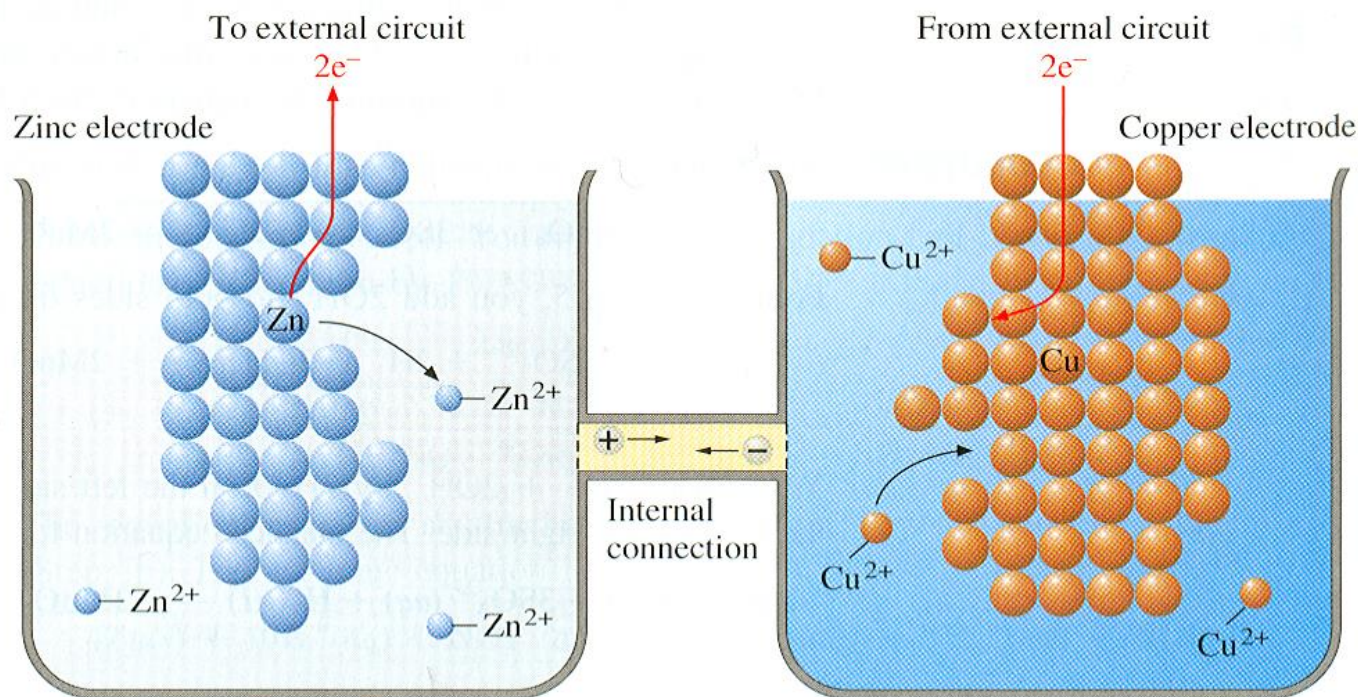


Pharmaceutical Analytical Chemistry II

PCC/PCD 105

Week 2 lecture

Oxidation Reduction Reactions



Electrode Potential

"Half-Cell Potential" "Single electrode potential"

The potential that develops at the phase boundary between
Electrode / Electrolyte solution

It is **impossible** to determine *absolute* half-cell potentials, we can readily determine *relative* electrode potentials, because all voltage-measuring devices measure only *differences* in potential.

Cell Potential

The *difference* between two half-cell potentials

$$E_{\text{cell}} = E_{\text{cathode}} - E_{\text{anode}}$$

"+ ... galvanic"

"- ... Electrolytic"

It is potential difference that develops between the Electrodes of the cell.

***Considered as a measure of the tendency for the reaction**

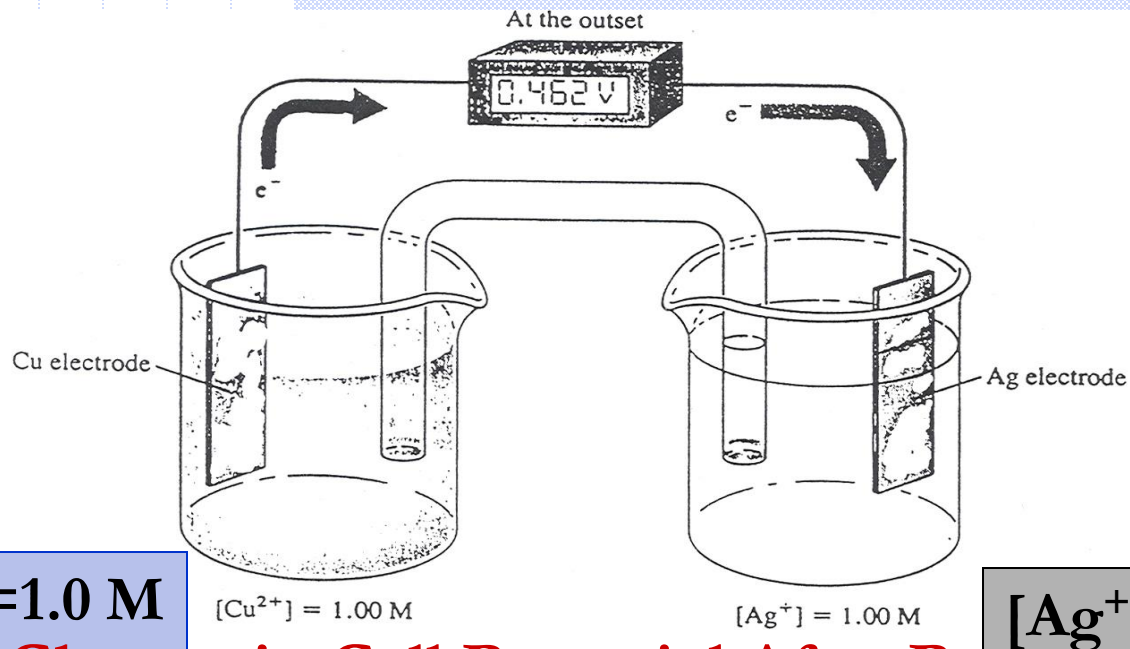
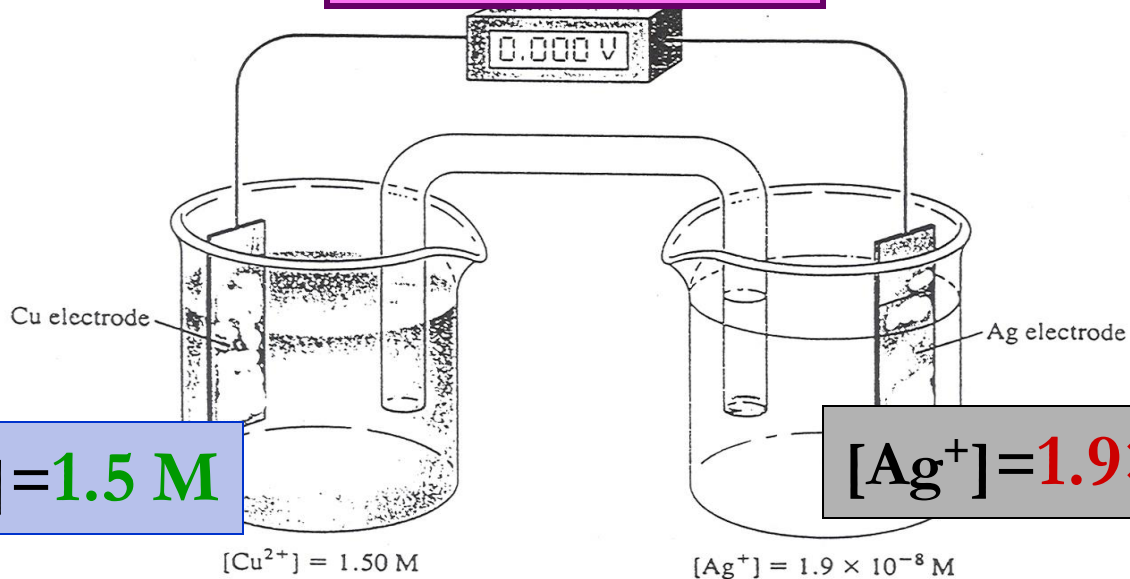


Figure 3: Change in Cell Potential After Passage of Current

AT EQUILIBRIUM



- ◆ According to Cell Rx: $2 \text{Ag}^+ + \text{Cu(s)} \rightleftharpoons 2 \text{Ag(s)} + \text{Cu}^{2+}$
- ◆ As shown in Figure-3(a), when the copper and silver ion activities in the cell are **1.00 M**, a potential of **0.462 V develops**, which shows that this reaction is far from equilibrium.
- ◆ As the reaction proceeds, this **potential becomes smaller** until at equilibrium the **meter reads 0.00 V**. As shown in Figure-3(b), the **copper** ion equilibrium concentration is then just slightly less than **1.5 M** (increased than the starting concentration) and the **silver** ion concentration is **$1.9 \times 10^{-8} \text{ M}$** “very low conc.”.
- ◆ At this point, no further net flow of electrons will occur.

◆ Although we cannot determine absolute potentials of electrodes, we **can** readily determine *relative* electrode potentials.

For example, if we replace the copper anode in Figure-1(a) with a cadmium electrode immersed in a cadmium sulfate solution, the voltmeter reads about 0.7 V greater than the original cell.

Define:

- ◆ Liquid junction potential
- ◆ Electrode potential
- ◆ Relative electrode potential
- ◆ Cell potential

Schematic representation

Cu/CuSO₄ Anode

Ag/AgNO₃ Cathode

Cu | Cu²⁺ (1.0 M) || Ag⁺ (1.0 M) | Ag

Cu | CuSO₄ (1.0 M) || AgNO₃ (1.0 M) | Ag

Example:

Zn/ZnSO₄ Anode

Cu/Cu SO₄ Cathode

Zn | Zn²⁺ (1.0 M) || Cu²⁺ (1.0 M) | Cu

Zn | ZnSO₄ (1.0 M) || CuSO₄ (1.0 M) | Cu

A single vertical line indicates a phase boundary or interface at which a potential develops "Electrode Potential"

Zn | ZnSO₄ Anode *potential: Potential develops at a phase boundary between Zinc electrode and Zinc sulfate solution*

A double vertical line represents the salt bridge

Anode is assigned as left-hand electrode

Cathode is assigned as right-hand electrode

() activity

TYPES of ELECTRODES "Half-Cell"

1. INERT Electrode:

It does not share in electrode reaction. It acts as a conductor of electrons .

Example: Pt, gold, graphite,... it is used in the following electrodes:

- Gas electrode,

E.g. Hydrogen electrode H_2 gas/ H^+



- Metal ions in different oxidation states Fe^{3+}/Fe^{2+}



- Non-Metal ions in different oxidation states

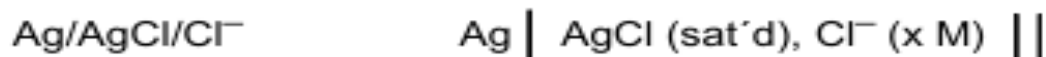


2. METAL / METAL ION Electrode,



3. METAL / INSOLUBLE SALT OF METAL / ION IN COMMON WITH SALT electrode

- Silver/silver chloride electrode:



- Calomel electrode:



Standard electrode is the electrode at standard conditions:

- 1- Conc. of ions in aq. solutions = 1M
- 2- Pressure of gases = 1 atm. (760 mm Hg)
- 3- temp. = 25° C (298 K)

E° .. Standard electrode potential..

Temperature dependent

N.B. E .. electrode potential (not standard)

The Standard Hydrogen Reference Electrode

- ◆ has been used throughout the world for many years as a **universal reference electrode**.
- ◆ It is a **typical gas electrode**.
- ◆ It is **reversible** and acts either as **an anode** or as a **cathode**, depending upon the half-cell with which it is coupled.

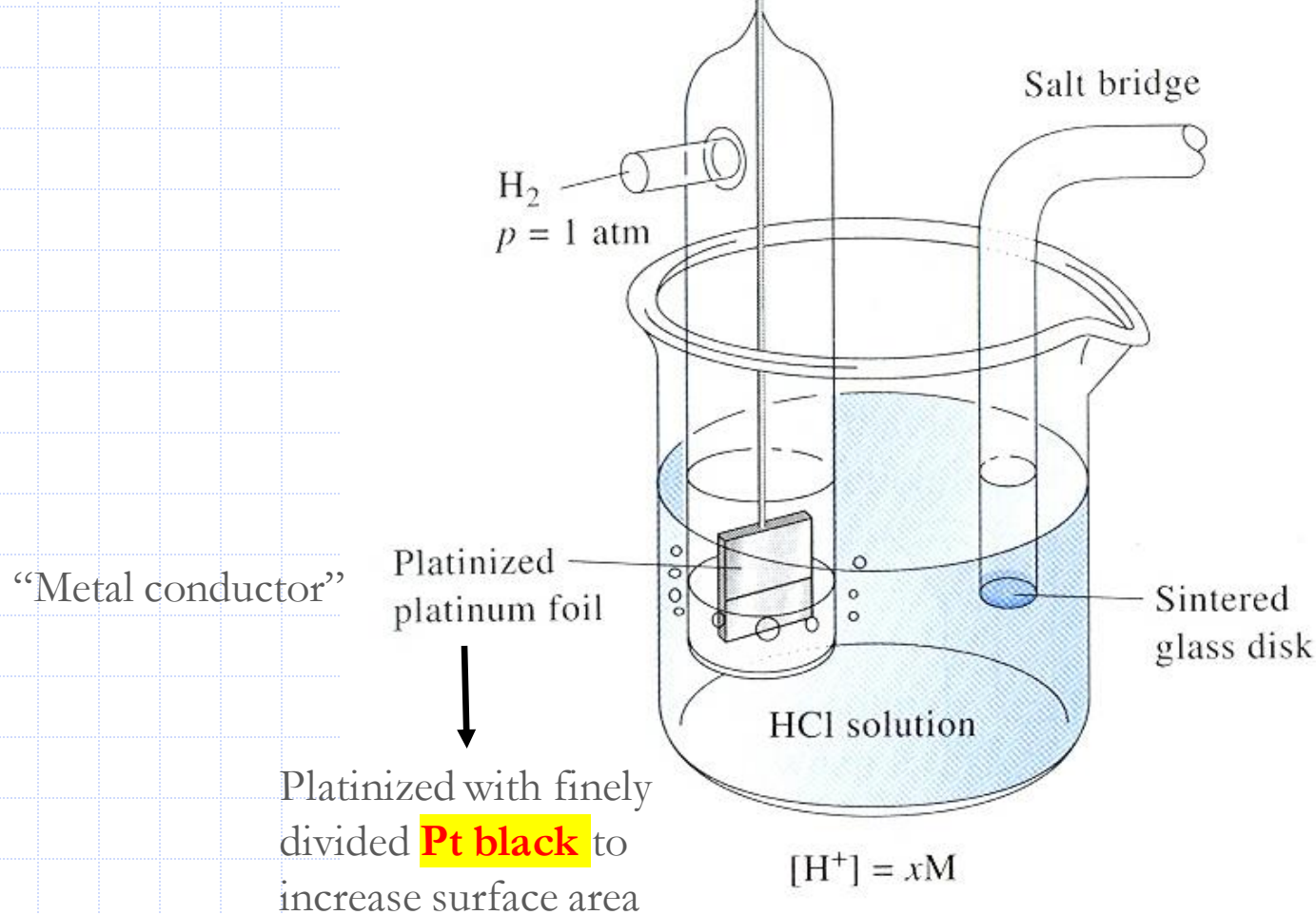


Hydrogen gas is oxidized to hydrogen ion when this electrode serves as the anode.

Hydrogen ion is reduced to molecular hydrogen when the hydrogen electrode acts as a cathode.

- ◆ *By convention, the potential of the standard hydrogen electrode is assigned a **value of zero** at all temperatures.*

Hydrogen Gas Electrode



Definition of Electrode Potential

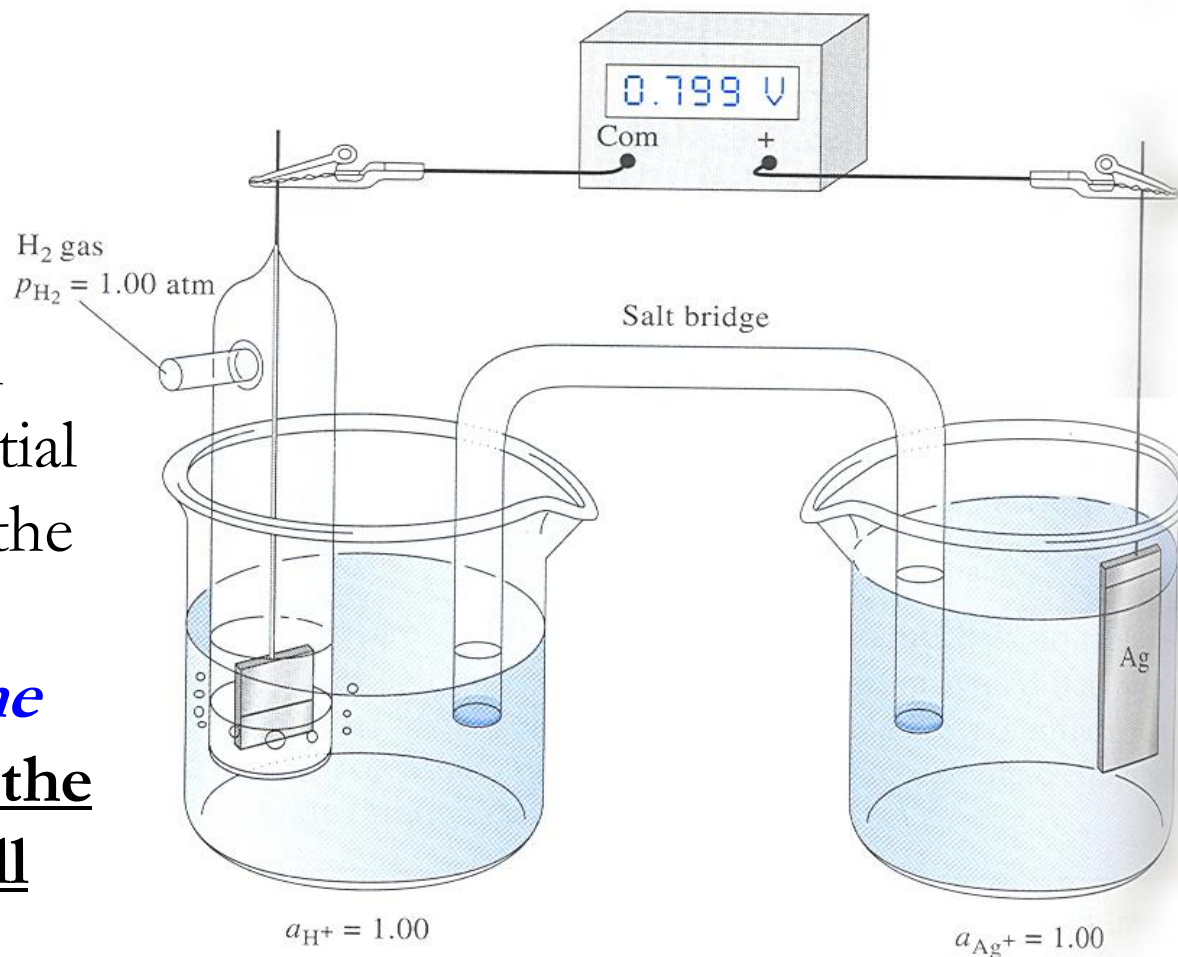
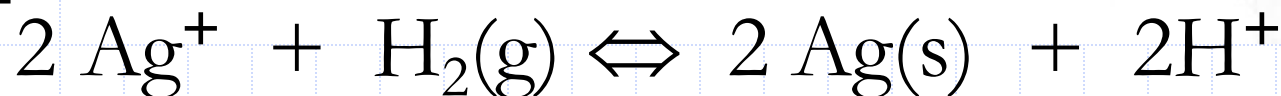
- ◆ An *electrode potential* is defined as the potential of a cell consisting of the electrode in question *acting as a cathode* and the standard hydrogen electrode *acting as an anode* “**relative electrode potential**”.

◆ For $\text{SHE} || \text{Ag}^+ (0.02 \text{ M}) | \text{Ag}$

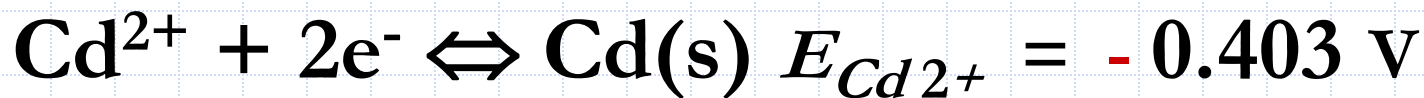
$$E_{\text{cell}} = E_{\text{Ag}} - 0.000 = E_{\text{Ag}}$$



This galvanic cell develops a potential of 0.799 V with the silver electrode functioning *as the cathode*; that is, the spontaneous cell reaction is:



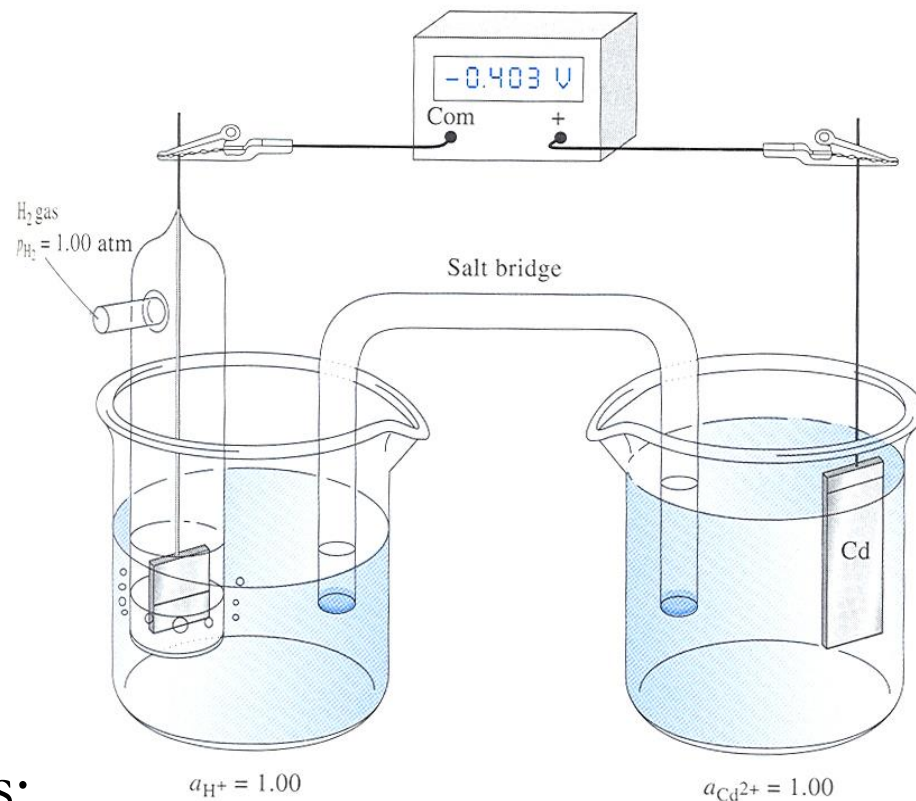
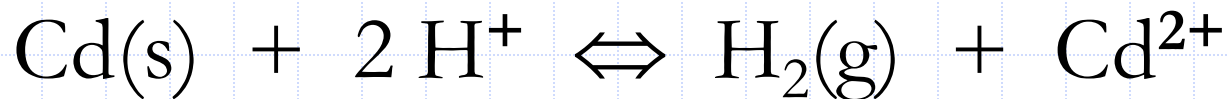
For



The **cadmium electrode** bears a **negative charge** with respect to the standard hydrogen electrode.

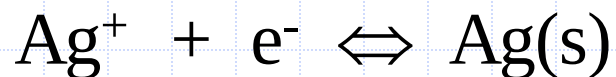
Zinc electrode functioning *as the anode*;

The **spontaneous** cell reaction is:

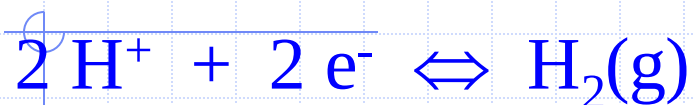


Half-Reaction

Electrode Potential, V



+ 0.799



0.000



- 0.403



- 0.763

The magnitudes of electrode potentials indicate the relative strengths of the ionic species as **electron acceptors (oxidizing agents)**; that is, in **decreasing** strength, undergoes reduction “i.e. as E increases, tendency for reduction increases, more powerful oxidizing agent”

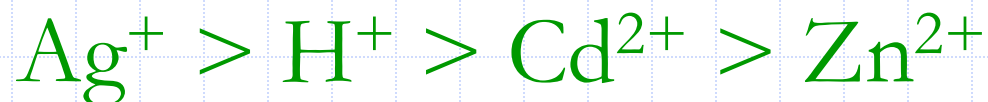


TABLE 18-1

Standard Electrode Potentials*

Reaction	E^0 at 25°C, V
$\text{Cl}_2(g) + 2e^- \rightleftharpoons 2\text{Cl}^-$	+1.359
$\text{O}_2(g) + 4\text{H}^+ + 4e^- \rightleftharpoons 2\text{H}_2\text{O}$	+1.229
$\text{Br}_2(aq) + 2e^- \rightleftharpoons 2\text{Br}^-$	+1.087
$\text{Br}_2(l) + 2e^- \rightleftharpoons 2\text{Br}^-$	+1.065
$\text{Ag}^+ + e^- \rightleftharpoons \text{Ag}(s)$	+0.799
$\text{Fe}^{3+} + e^- \rightleftharpoons \text{Fe}^{2+}$	+0.771
$\text{I}_3^- + 2e^- \rightleftharpoons 3\text{I}^-$	+0.536
$\text{Cu}^{2+} + 2e^- \rightleftharpoons \text{Cu}(s)$	+0.337
$\text{UO}_2^{2+} + 4\text{H}^+ + 2e^- \rightleftharpoons \text{U}^{4+} + 2\text{H}_2\text{O}$	+0.334
$\text{Hg}_2\text{Cl}_2(s) + 2e^- \rightleftharpoons 2\text{Hg}(l) + 2\text{Cl}^-$	+0.268
$\text{AgCl}(s) + e^- \rightleftharpoons \text{Ag}(s) + \text{Cl}^-$	+0.222
$\text{Ag}(\text{S}_2\text{O}_3)_2^{3-} + e^- \rightleftharpoons \text{Ag}(s) + 2\text{S}_2\text{O}_3^{2-}$	+0.017
$2\text{H}^+ + 2e^- \rightleftharpoons \text{H}_2(g)$	0.000
$\text{AgI}(s) + e^- \rightleftharpoons \text{Ag}(s) + \text{I}^-$	-0.151
$\text{PbSO}_4 + 2e^- \rightleftharpoons \text{Pb}(s) + \text{SO}_4^{2-}$	-0.350
$\text{Cd}^{2+} + 2e^- \rightleftharpoons \text{Cd}(s)$	-0.403
$\text{Zn}^{2+} + 2e^- \rightleftharpoons \text{Zn}(s)$	-0.763

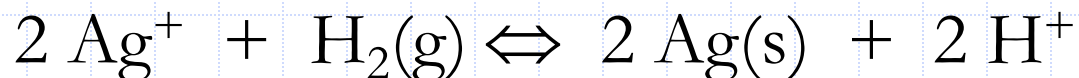
*See Appendix 5 for a more extensive list.

The sign Convention for Electrode Potentials

- ◆ The **sign of an electrode** potential is determined by the sign of the half-cell in question when it is coupled to a standard hydrogen electrode.
- ◆ When the half-cell of interest behaves **spontaneously as a cathode**, it is the **positive electrode** of the galvanic cell. Thus, its electrode potential is **positive**.
- ◆ When the half-cell of interest behaves **as an anode**, the electrode is **negative**. Thus, its electrode potential is **negative**.
- ◆ It is important to emphasize that the electrode potential refers to a **half-cell process** written *as a reduction*.
- ◆ For the **zinc** and **cadmium** electrodes, the spontaneous reactions are oxidations. (spontaneous anodes)

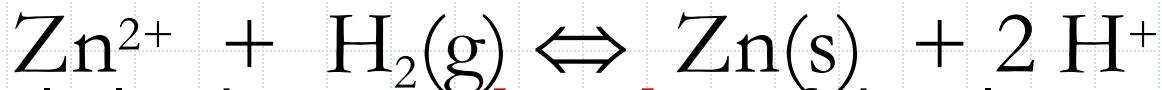
◆ It is evident, then, that the sign of an electrode potential indicates **whether the reduction is spontaneous** with respect to the **standard hydrogen electrode**.

◆ The **positive sign** associated with the electrode potential for silver indicates that the process:



favors the products under ordinary conditions.

Ag electrode is spontaneous cathode



◆ Similarly, the **negative sign** of the electrode potential for zinc means that the analogous reaction, is **not spontaneous**. That is, the reaction tends to go in the other direction:



TABLE 18-1

Standard Electrode Potentials*

Reaction	E^0 at 25°C, V
$\text{Cl}_2(g) + 2e^- \rightleftharpoons 2\text{Cl}^-$	+1.359
$\text{O}_2(g) + 4\text{H}^+ + 4e^- \rightleftharpoons 2\text{H}_2\text{O}$	+1.229
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$\text{Ag}^+ + e^- \rightleftharpoons \text{Ag}(s)$	+ 0.799
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$\text{Ag}(\text{S}_2\text{O}_3)_2^{3-} + e^- \rightleftharpoons \text{Ag}(s) + 2\text{S}_2\text{O}_3^{2-}$	+ 0.017
$2\text{H}^+ + 2e^- \rightleftharpoons \text{H}_2(g)$	0.000
$\text{AgI}(s) + e^- \rightleftharpoons \text{Ag}(s) + \text{I}^-$	- 0.151
$\text{PbSO}_4 + 2e^- \rightleftharpoons \text{Pb}(s) + \text{SO}_4^{2-}$	- 0.350
$\text{Cd}^{2+} + 2e^- \rightleftharpoons \text{Cd}(s)$	- 0.403
$\text{Zn}^{2+} + 2e^- \rightleftharpoons \text{Zn}(s)$	- 0.763

*See Appendix 5 for a more extensive list.

◆ Consider the half-reaction:

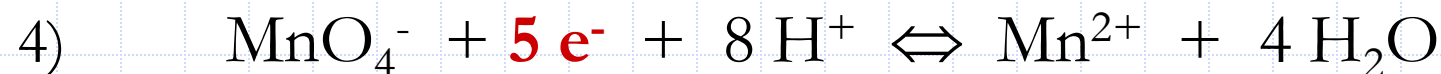
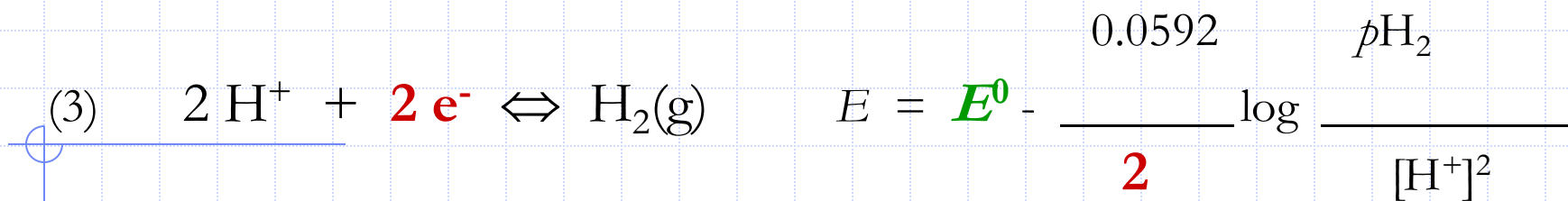


$$E = \textcolor{green}{E^0} - \frac{0.0592}{\textcolor{red}{n}} \log \frac{[C]^c [D]^d \dots}{[A]^a [B]^b \dots} = \frac{[\text{Products}]}{[\text{Reactants}]}$$

◆ The above Equation is known as the
Nernst equation

$$(1) \quad \text{Zn}^{2+} + \textcolor{red}{2 e^-} \rightleftharpoons \text{Zn(s)} \quad E = \textcolor{green}{E^0} - \frac{0.0592}{\textcolor{red}{2}} \log \frac{1}{[\text{Zn}^{2+}]}$$

$$(2) \quad \text{Fe}^{3+} + \textcolor{red}{e^-} \rightleftharpoons \text{Fe}^{2+} \quad E = \textcolor{green}{E^0} - \frac{0.0592}{\textcolor{red}{1}} \log \frac{[\text{Fe}^{2+}]}{[\text{Fe}^{3+}]}$$



$$E = E^0 - \frac{0.0592}{5} \log \frac{[\text{Mn}^{2+}]}{[\text{MnO}_4^-][\text{H}^+]^8}$$

Cell Potential Calculations

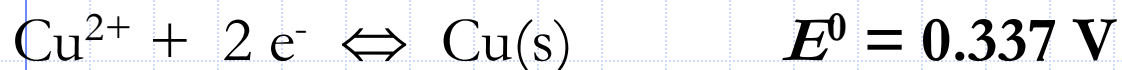
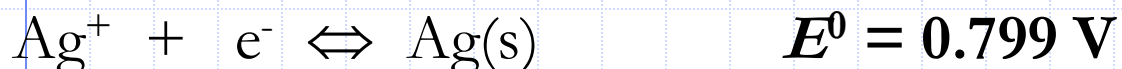
$$E_{\text{cell}} = E_{\text{cathode}} - E_{\text{anode}}$$

Example-1:

Calculate the thermodynamic potential of the following cell:



The two half-reactions and standard potentials are



The electrode potentials are,

$$E_{\text{Ag}^{+}} = 0.799 - 0.0592 \log \frac{1}{0.200} = 0.758 \text{ V}$$

$$E_{\text{Cu}^{2+}} = 0.337 - \frac{0.0592}{2} \log \frac{1}{0.100} = 0.307 \text{ V}$$

$$E_{\text{cell}} = E_{\text{Ag}^{+}} - E_{\text{Cu}^{2+}} = 0.758 - 0.307 = + 0.451$$

Galvanic Cell-Spontaneous Rx

Example -2:

Calculate the thermodynamic potential for the cell



The electrode potentials for the two half-reactions are identical to the electrode potentials calculated in Example-4.

That is,

$$E_{\text{Ag}^+} = 0.758 \text{ V} \quad \text{and} \quad E_{\text{Cu}^{2+}} = 0.307 \text{ V}$$

In contrast to the previous example, however, the silver electrode is the **anode** and the copper electrode the **cathode**.

Substituting these electrode potentials into Equation-10 gives:

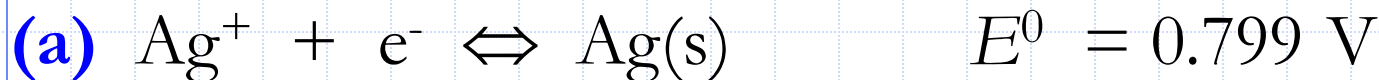
$$E_{\text{cell}} = E_{\text{Cu}^{2+}} - E_{\text{Ag}^+} = 0.307 - 0.758 = -0.451 \text{ V}$$

Electrolytic Cell- non Spontaneous Rx

Example-3:

Calculate the potential of a silver electrode immersed in a 0.0500 M solution of NaCl and saturated AgCl(s) using,

(a) $E^0_{\text{Ag}^+} = 0.799 \text{ V}$ and (b) $E^0_{\text{AgCl}} = 0.222 \text{ V}$.

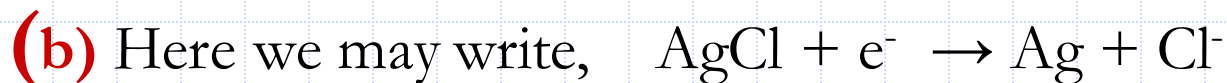


The Ag^+ concentration of this solution is given by

$$[\text{Ag}^+] = K_{\text{sp}}/[\text{Cl}^-] = 1.82 \times 10^{-10}/0.0500 = 3.64 \times 10^{-9} \text{ M}$$

Substituting into the Nernst expression gives

$$E = 0.799 - 0.0592 \log \frac{1}{3.64 \times 10^{-9}} = 0.299 \text{ V}$$



$$E = 0.222 - 0.0592 \log [\text{Cl}^-] = 0.222 - 0.0592 \log 0.0500 = 0.299 \text{ V}$$